

Practical Insights from Machine Learning

The Big Data Newsvendor

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The Newsvendor Problem

A company sells perishable goods and needs to make an order before observing the uncertain demand. For repetitive sales, a sensible goal is to order a quantity that minimises the total expected cost according to

$$\min_{q \geq 0} EC(q) := \mathbb{E}[C(q; D)],$$

where q is the order quantity, $D \in \mathcal{D}$ is the uncertain (random) future demand,

$$C(q; D) := b(D - q)^+ + h(q - D)^+$$

is the random cost of order q and demand D , and b and h are, respectively, the unit back-ordering and holding costs.



The Data-Driven Newsvendor Problem

- If the demand distribution, F , is known,

$$q^* = \inf \left\{ y : F(y) \geq \frac{b}{b+h} \right\}.$$

- In practice, the true distribution is unknown. But with historical demand observations $\mathbf{d}(n) = [d_1, \dots, d_n]$, a sample average approximation (SAA) approach is to substitute the true expectation with a sample average expectation and solve:

$$\min_{q \geq 0} \hat{R}(q; \mathbf{d}(n)) = \frac{1}{n} \sum_{i=1}^n [b(d_i - q)^+ + h(q - d_i)^+].$$

The optimal SAA decision is given by

$$\hat{q}_n = \inf \left\{ y : \hat{F}_n(y) \geq \frac{b}{b+h} \right\},$$

where $\hat{F}(\cdot)$ is the empirical cumulative distribution function of the demand from the n observations.

The Feature-Based Newsvendor Problem

In practice, the demand depends on many observable *features*, such as seasonality, weather, location, and economic indicators, which are available prior to making the order.

In other words, the real newsvendor problem is to optimise the conditional expected cost function based on the feature vector $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^p$:

$$\min_{\substack{q(\cdot) \in \mathcal{L} \\ q: \mathcal{X} \rightarrow \mathbb{R}}} \mathbb{E}[C(q(\mathbf{x}); D(\mathbf{x})) \mid \mathbf{x}].$$

Three Issues to Address

- What features does the demand depend?

We assume that the decision maker (DM) has already collected appropriate historical data $S_n = [(\mathbf{x}_1, d_1), \dots, (\mathbf{x}_n, d_n)]$. The data may be low dimensional or high dimensional.

- How to solve the problem in an efficient manner given the feature-demand data set?

The ERM and KO approaches.

- What performance guarantee is possible prior to observing the demand in the next period?

ERM Algorithm

The Empirical Risk Minimisation (ERM) approach to solving the newsvendor problem with feature data is

$$\min_{\substack{q(\cdot) \in \mathcal{L} \\ q: \mathcal{X} \rightarrow \mathbb{R}}} \hat{R}(q(\cdot); S_n) = \frac{1}{n} \sum_{i=1}^n [b(d_i - q(\mathbf{x}_i))^+ + h(q(\mathbf{x}_i) - d_i)^+].$$

(NV-ERM)

Consider linear decision rules of the form

$$\mathcal{L} = \left\{ q: \mathcal{X} \rightarrow \mathbb{R} : q(\mathbf{x}) = \mathbf{q}'\mathbf{x} = \sum_{j=1}^p q^j x^j \right\},$$

where $x^1 = 1$ to allow for a feature-independent term (an intercept term).

When p is relatively small, the DM can solve (NV-ERM) via:

$$\begin{aligned} \min_{q: q(\mathbf{x}) = \sum_{j=1}^p q^j x^j} \hat{R}(q(\cdot); S_n) &= \frac{1}{n} \sum_{i=1}^n [b(d_i - q(\mathbf{x}_i))^+ + h(q(\mathbf{x}_i) - d_i)^+] \\ &\equiv \min_{\mathbf{q}=[q^1, \dots, q^p]} \frac{1}{n} \sum_{i=1}^n (bu_i + ho_i), \\ \text{s.t. } \forall i = 1, \dots, n: \quad u_i &\geq d_i - q^1 - \sum_{j=2}^p q^j x_i^j, \\ o_i &\geq q^1 + \sum_{j=2}^p q^j x_i^j - d_i, \\ u_i, o_i &\geq 0, \end{aligned} \tag{NV-ERM1}$$

where the dummy variables u_i and o_i represent, respectively, under-age and overage costs in period i .

(NV-ERM1): algorithmically stable but loose for performance guarantee when p is large. In the case of high dimensional data, one can automate feature selection by solving the following regularised version of (NV-ERM1):

$$\begin{aligned} & \min_{q: q(\mathbf{x}) = \sum_{j=1}^p q^j x^j} \hat{R}(q(\cdot); S_n) + \lambda \|\mathbf{q}\|_k \\ &= \frac{1}{n} \sum_{i=1}^n [b(d_i - q(\mathbf{x}_i))^+ + h(q(\mathbf{x}_i) - d_i)^+] + \lambda \|\mathbf{q}\|_k \\ &\equiv \min_{\mathbf{q}=[q^1, \dots, q^p]} \frac{1}{n} \sum_{i=1}^n (bu_i + ho_i) + \lambda \|\mathbf{q}\|_k, \end{aligned}$$

$$\text{s.t. } \forall i = 1, \dots, n: \quad u_i \geq d_i - q^1 - \sum_{j=2}^p q^j x_i^j,$$

$$o_i \geq q^1 + \sum_{j=2}^p q^j x_i^j - d_i,$$

$$u_i, o_i \geq 0.$$

(NV-ERM2)

Nadaraya-Watson Kernel Regression

Given past data $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$, one want to estimate

$$m(\mathbf{x}_{n+1}) = \mathbb{E}[Y \mid \mathbf{x}_{n+1}],$$

where $Y \in \mathbb{R}$ is the dependent variable and $\mathbf{x}_{n+1} \in \mathbb{R}^p$ is a vector of independent variables. Instead, we estimate by the locally weighted average

$$m_h(\mathbf{x}_{n+1}) = \frac{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i) y_i}{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i)},$$

where $K_w(\cdot) = K(\cdot/w)/w$ is a kernel function with bandwidth w .

Typical example:

- Uniform kernel: $K(\mathbf{u}) = \frac{1}{2} \mathbb{1}(\|\mathbf{u}\|_2 \leq 1)$;
- Gaussian kernel: $K(\mathbf{u}) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\|\mathbf{u}\|_2^2}{2}\right)$.

KO Method

Based on Nadaraya-Watson regression, for the feature-dependent newsvendor expected cost $\mathbb{E}[C(q; D) \mid \mathbf{x}_{n+1}]$, we can estimate

$$\frac{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i) C(q, d_i)}{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i)}.$$

The Kernel Optimization (KO) Method is

$$\min_{q \geq 0} \tilde{R}(q; S_n, \mathbf{x}_{n+1}) = \min_{q \geq 0} \frac{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i) C(q, d_i)}{\sum_{i=1}^n K_w(\mathbf{x}_{n+1} - \mathbf{x}_i)}. \quad (\text{NV-KO})$$

KO Method (Cont.)

Proposition

The optimal feature-based newsvendor decision $\hat{q}_n^\kappa$ obtained by solving (NV-KO) is given by

$$\hat{q}_n^\kappa = \hat{q}_n^\kappa(\mathbf{x}_{n+1}) = \inf \left\{ q : \frac{\sum_{i=1}^n \kappa_i \mathbb{1}(d_i \leq q)}{\sum_{i=1}^n \kappa_i} \geq \frac{b}{b+h} \right\},$$

where for simplicity we introduce $\kappa_i = K_w(\mathbf{x}_{n+1} - \mathbf{x}_i)$. In other words, we can find $\hat{q}_n^\kappa$ by ranking the past demand in increasing order, and choosing the smallest value at which the inequality is satisfied.

Expected Cost Difference

Theorem

Let $q^ = q^*(\tilde{x})$ denote the true optimal newsvendor decision given feature $\tilde{x}$ and $\hat{q}$, some other decision not equal to q^* . Then the difference of the expected costs of the two decisions is given by*

$$\mathbb{E}C(\hat{q}; D) - \mathbb{E}C(q^*; D) = (b + h)\mathbb{E}[|\hat{q} - D| \mathbf{1}\{(\hat{q} \wedge q^*) \leq D \leq (\hat{q} \vee q^*)\}]$$

Proof. Let $\underline{D} \leq q_1 \leq q_2 \leq \bar{D}$, then consider the difference $C(q_2; \cdot) - C(q_1; \cdot)$ for three regions: $D < q_1$, $q_1 \leq D \leq q_2$ and $D > q_2$,

$$\begin{aligned}
 & C(q_2; D) - C(q_1; D) \\
 &= -b(q_2 - q_1)\mathbb{1}(D > q_2) + h(q_2 - q_1)\mathbb{1}(D < q_1) \\
 &\quad + [bq_1 + hq_2 - (b + h)D]\mathbb{1}(q_1 \leq D \leq q_2) \\
 &= -b(q_2 - q_1)\mathbb{1}(D > q_2) + h(q_2 - q_1)\mathbb{1}(D < q_1) \\
 &\quad + [b(q_2 - (q_2 - q_1)) + hq_2 - (b + h)D]\mathbb{1}(q_1 \leq D \leq q_2) \\
 &= -b(q_2 - q_1)\mathbb{1}(D > q_1) + h(q_2 - q_1)\mathbb{1}(D < q_1) \\
 &\quad + (b + h)(q_2 - D)\mathbb{1}(q_1 \leq D \leq q_2).
 \end{aligned} \tag{1}$$

Similarly,

$$\begin{aligned}
 & C(q_2; D) - C(q_1; D) \\
 &= -b(q_2 - q_1)\mathbb{1}(D > q_2) + h(q_2 - q_1)\mathbb{1}(D < q_1) \\
 &\quad + [bq_1 + h(q_1 + (q_2 - q_1)) - (b + h)D]\mathbb{1}(q_1 \leq D \leq q_2) \\
 &= -b(q_2 - q_1)\mathbb{1}(D > q_2) + h(q_2 - q_1)\mathbb{1}(D < q_2) \\
 &\quad - (b + h)(D - q_1)\mathbb{1}(q_1 \leq D \leq q_2).
 \end{aligned} \tag{2}$$

Case I: Let $q_1 = q^*$ such that $\mathbb{P}(D < q_1) = \frac{b}{b+h}$ and $\mathbb{P}(D > q_1) = \frac{h}{b+h}$, then the first two terms in (1) are 0, so

$$|\mathbb{E}C(q_2; D) - \mathbb{E}C(q_1; D)| = (b+h)\mathbb{E}[(q_2 - D)\mathbb{1}(q_1 \leq D \leq q_2)].$$

Case II: Let $q_2 = q^*$ such that $\mathbb{P}(D < q_2) = \frac{b}{b+h}$ and $\mathbb{P}(D > q_2) = \frac{h}{b+h}$, then the first two terms in (2) are 0, so

$$|\mathbb{E}C(q_2; D) - \mathbb{E}C(q_1; D)| = (b+h)\mathbb{E}[(D - q_1)\mathbb{1}(q_1 \leq D \leq q_2)].$$

In sum, the expected cost difference between q^* and $\hat{q}$ increases as $|q^* - \hat{q}|$ increases. $\square$

Generalisation error

The performance bound splits into two components: **finite-sample bias** (arise from having a finite amount of data) and **generalisation error** (the generalisability of the in-sample decision to out-of-sample data).

The generalisation error provides a description of the degree of over-fitting for the DM's decision.

Out-of-Sample Cost

Definition (True Risk).

The **true risk** is the expected out-of-sample cost, with which the expectation is taken over an **unknown distribution** over $\mathcal{X} \times \mathcal{D}$, where $\mathcal{X} \subset \mathbb{R}^p$:

$$R_{\text{true}}(q) := \mathbb{E}_{D(\mathbf{x})}[C(q; D(\mathbf{x}))].$$

We cannot minimise this cost because the distribution is unknown. The **empirical risk** is the average cost over $\mathcal{X} \times \mathcal{D}$, where $\mathcal{X} \subset \mathbb{R}^p$:

$$\hat{R}(q; S_n) := \frac{1}{n} \sum_{i=1}^n C(q, d_i(\mathbf{x}_i)),$$

which can be calculated.

Assumptions

Assumption 1.

The feature vector $\mathbf{X}$ is normalised ($\mathbf{X}_1 = 1$ a.s., $\mathbf{X}_{[2:p]}$ has mean zero and SD one), and it lives in a closed unit ball:

$$\|\mathbf{X}\| \leq X_{\max} \sqrt{p}.$$

To simplify the exposition, and the results do not require that the DM knows the true mean or SD of $\mathbf{X}$, only the size bound $X_{\max}$.

Assumption 2.

The demand follows the linear model

$$D \mid (\mathbf{X} = \mathbf{x}) = \beta^T \mathbf{x} + \varepsilon$$

where the distribution $\varepsilon, f_\varepsilon$, is bounded away from zero on the domain $[\underline{D}, \bar{D}]$ (otherwise unspecified).

Assumption 3.

All decision functions (policies) described are measurable, and $\mathcal{L}$ is a convex subset of a linear space.

Out-of-Sample Performance of NV-ERM1

Denote the true optimal solution by $q^* = q^*(\mathbf{x}_{n+1})$ and the decision resulting from (NV-ERM1) by $\hat{q} = \hat{q}(\mathbf{x}_{n+1})$. Then, with probability at least $1 - \delta$ over the random draw of the sample S_n , where each element of S_n is drawn *i.i.d.* from an unknown distribution on $\mathcal{X} \times \mathcal{D}$, and for all $n \geq 3$,

$$\begin{aligned} & \left| R_{\text{true}}(q^*) - \hat{R}_{\text{in}}(\hat{q}, S_n) \right| \\ & \leq (b \vee h) \bar{D} \left[\frac{2(b \vee h)}{b \wedge h} \cdot \frac{p}{n} + \left(\frac{4(b \vee h)}{b \wedge h} \cdot p + 1 \right) \sqrt{\frac{\log(\frac{2}{\delta})}{2n}} \right] \\ & \quad + (b \vee h) K \frac{\sqrt{\log n}}{n^{\frac{1}{2+p}}}, \end{aligned}$$

where

$$K = \sqrt{\frac{9(8+5p)}{4+p}} \cdot \frac{1}{\left(1 - 2^{-\frac{4}{4+p}}\right) \lambda_2^*}, \quad \lambda_2^* = \min_{t \in [\underline{D}, \bar{D}]} f_\varepsilon(t).$$

Remarks.

- This is a statement about how close the in-sample cost of the **in-sample decision** $R_{\text{in}}(\hat{q}, S_n)$ is to the expected cost of the **true optimal decision** $R_{\text{true}}(q^*)$.
- The first term on the RHS is the bound on the **generalisation error**. For fixed cost parameters b and h , we find it scales as $O\left(\frac{p}{\sqrt{n}}\right)$: if $\frac{p}{\sqrt{n}}$ is large, or growing, then overfitting will be an issue, and NV-ERM1 may not be a good algorithm.
- The second term on the RHS is the **finite-sample bias**, which could only be reduced by collecting more observations.

Out-of-Sample Performance of NV-ERM2

Same setting as (NV-ERM1), and we denote the decision resulting from (NV-ERM2) by $\hat{q}_\lambda = \hat{q}_\lambda(\mathbf{x}_{n+1})$. Then, with probability at least $1 - \delta$ over the random draw of the sample S_n , where each element of S_n is drawn *i.i.d.* from an unknown distribution on $\mathcal{X} \times \mathcal{D}$, and for all $n \geq 3$,

$$\begin{aligned} & \left| R_{\text{true}}(q^*) - \hat{R}_{\text{in}}(\hat{q}_\lambda, S_n) \right| \\ & \leq (b \vee h) \bar{D} \left[\frac{(b \vee h) X_{\max}^2}{\lambda \bar{D}} \cdot \frac{p}{n} + \left(\frac{2(b \vee h) X_{\max}^2}{\lambda \bar{D}} \cdot p + 1 \right) \sqrt{\frac{\log(\frac{2}{\delta})}{2n}} \right] \\ & \quad + (b \vee h) K \frac{\sqrt{\log n}}{n^{\frac{1}{2+\frac{p}{4}}}} + (b \vee h) \mathbb{E}_{D|\mathbf{x}_{n+1}}[|\hat{q}_\lambda - \hat{q}|], \end{aligned}$$

where

$$K = \sqrt{\frac{9(8+5p)}{4+p}} \cdot \frac{1}{\left(1 - 2^{-\frac{4}{4+p}}\right) \lambda_2^*}, \quad \lambda_2^* = \min_{t \in [\underline{D}, \bar{D}]} f_\epsilon(t).$$

Remarks.

- The first term on the RHS is the bound on the **generalisation error** scales as $O\left(\frac{p}{\sqrt{n\lambda}}\right)$, which is smaller as λ larger. Choosing $\lambda = O\left(\frac{1}{p^2}\right)$ retrieves the same error rate as for NV-ERM1.
- The second term on the RHS is the **finite-sample bias**, which could only be reduced by collecting more observations.
- The third term on the RHS is the **regularisation bias** (the bias from the optimisation problem and the true problem), which is larger as λ larger.

Out-of-Sample Performance of NV-KO

Same setting as (NV-ERM1), and we denote decision to (NV-KO) with the Gaussian kernel by $\hat{q}^\kappa = \hat{q}^\kappa(\mathbf{x}_{n+1})$. Then,

$$\begin{aligned} \left| R_{\text{true}}(q^*) - \hat{R}_{\text{in}}(\hat{q}^\kappa, S_n) \right| &\leq (b \vee h) \bar{D} \cdot \\ &\left[\frac{2(b \vee h)}{b \vee h} \cdot \frac{1}{1 + (n-1)r_w(p)} + \left(\frac{4(b \vee h)}{\frac{1}{n} + (1 - \frac{1}{n})r_w(p)} + 1 \right) \sqrt{\frac{\log(\frac{2}{\delta})}{2n}} \right] \\ &+ (b \vee h) K \frac{\sqrt{\log n}}{n^{\frac{1}{2+\frac{p}{2}}}} + (b \vee h) \mathbb{E}_{D|\mathbf{x}_{n+1}} [|\hat{q}^\kappa - \hat{q}|], \end{aligned}$$

where

$$r_w(p) = \exp\left(-\frac{2X_{\max}^2 p}{w^2}\right),$$

w is the kernel bandwidth, and

$$K = \sqrt{\frac{9(8+5p)}{4+p}} \cdot \frac{1}{\left(1 - 2^{-\frac{4}{4+p}}\right) \lambda_2^*}, \quad \lambda_2^* = \min_{t \in [\underline{D}, \bar{D}]} f_\varepsilon(t).$$

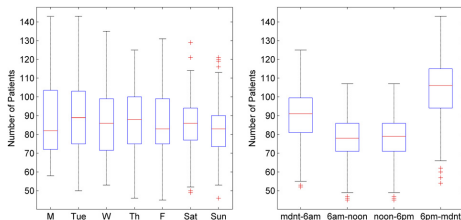
Remarks.

- The first term on the RHS is the bound on the **generalisation error** which scales as $O\left(\frac{1}{r_w(p)\sqrt{n}}\right)$, which can be controlled by increasing the kernel bandwidth w .
- The second term on the RHS is the **finite-sample bias**, which could only be reduced by collecting more observations.
- The third term on the RHS is the **regularisation bias**, which is larger as w larger.

Nurse Staffing in a Hospital Emergency Room

We can approximate the nurse-staffing problem as a newsvendor problem. We assume the hospital incurs a linear underage cost if too many patients arrive and expensive **agency nurses** have to be called, and a linear overage cost if too many **regular nurses** are scheduled compared with the number of patients.

The data comes from the emergency room of a large teaching hospital in the UK from July 2008 to June 2009. We assume the hourly wage of an agency nurse is 2.5 times that of a regular nurse.



Methods Considered

- Solve (NV-ERM1) with the day of the week and time of the day features and an increasing number of days of past demands (for up to two weeks) with and without operational statistics features.
- Solve (NV-ERM2) with the day of the week and time of the day features and two weeks of past demands with and without OS features for a range of regularisation parameters.
- Solve (NV-KO) with the Gaussian kernel with the day of the week and time of the day features and an increasing number of days of past demands (for up to two weeks) with and without OS features.

Result

Method	Calibrated parameter	Avg. computation time (per iteration)	Mean (95 % CI)	% savings relative to SAA-day	Annual cost savings rel. to SAA-day
1a. SAA-day	—	14.0 s	1.523 ($\pm$ 0.109)	—	—
1b. Cluster + SAA	—	14.9 s	1.424 ($\pm$ 0.102)	—	—
2a. Ker-0	$w = 0.08$	0.0444 s	1.208 ($\pm$ 0.146)	20.7%	£39,915 (\$63,864)
2b. Ker-OS	$w = 1.62$	0.0494 s	1.156 ($\pm$ 0.140)	24.1%	£46,555 (\$74,488)
3a. NV-0	12 days	325 s	1.326 ($\pm$ 0.100)	12.9%	£24,909 (\$39,854)
3b. NV-OS	Four days	360 s	1.463 ($\pm$ 0.144)	—	—
4a. NVreg1	1×10^{-7}	84.5 s	1.336 ($\pm$ 0.100)	—	—
4b. NVreg1-OS	1×10^{-7}	114 s	1.174 ($\pm$ 0.113)	22.9%	£44,219 (\$70,750)
5a. NVreg2	5×10^{-7}	79.6 s	1.336 ($\pm$ 0.110)	—	—
5b. NVreg2-OS	1×10^{-7}	107 s	1.215 ($\pm$ 0.111)	20.2%	£39,065 (\$62,503)

- All the methods yield cost improvement. The best method: (NV-KO) with operational statistics features for $w = 1.62$. The second-best method, (NV-ERM1) with operational statistics features for $\lambda = 1 \times 10^{-7}$.
- The computational costs of the KO method is three orders of magnitude faster than the ERM-based methods.

Reference & Discussions

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